

Receipt Redaction Bounds: Contraction Coefficients for Publishable Privacy Artifacts

Draft v0.10 (2026-02-23)

Abstract

Several papers in this series treat anonymity as a *publishable artifact*: a receipt, certificate, or transcript that third parties can verify. Publishing such artifacts creates a new channel: the receipt itself can leak information unless it is carefully redacted.

We give a compact theory module for bounding that incremental leak. We model a redaction as a channel (Markov kernel) from sensitive internal state to published fields and bound the induced total-variation (TV) distinguishability in terms of a *contraction coefficient*. We highlight a practical corollary: if a published field is produced by an ε -local-DP (LDP) mechanism, then its TV contribution is at most $\tanh(\varepsilon/2)$, independent of the underlying state distribution. This yields a clean, composable knob for “how much you can publish” when attaching receipts to releases, audits, and monitoring logs.

Scope. We give small, reusable lemmas about *receipt redaction*: how much information can be removed from a public release ledger while still preserving the ability to verify key claims.

Imports. We import the notion of “release receipts” and their use as public commitments (Release A), and the broader ABOM/OINL interface contract (Anonymity C).[2, 3]

Exports. Exports two redaction bounds: (i) conditions under which removing a field preserves the ability to recompute a verifier hash, and (ii) a compositional rule for combining redactions across nested receipt layers (e.g., per-build receipts inside a transparency log).

Artifact policy. This archive is source-only; supporting artifacts are available on request as a bundle committed by ABOM/OINL and governed by the series artifact policy (Synthesis 6).[1]

1 Receipt fields as a channel

Let X denote a sensitive internal object (e.g., a raw trace, a routing transcript, a congestion time series, or a labeled audit episode), and let R denote a published receipt field (or a tuple of fields). A *redaction* is a (possibly randomized) mapping from X to R .

Definition 1 (Receipt channel and Dobrushin coefficient). *A receipt channel is a Markov kernel $K(\cdot \mid x)$ from $\mathcal{X}$ to $\mathcal{R}$. Its (TV) contraction coefficient is*

$$\eta(K) := \sup_{x, x'} \text{TV}(K(\cdot \mid x), K(\cdot \mid x')) \in [0, 1].$$

Lemma 1 (TV cannot increase through a channel). *For any two distributions P, Q on $\mathcal{X}$ and any channel K ,*

$$\text{TV}(PK, QK) \leq \eta(K) \text{TV}(P, Q).$$

In particular, $\text{TV}(PK, QK) \leq \text{TV}(P, Q)$.

Proof. This is a standard Dobrushin-contraction bound. One proof uses the dual characterization of TV as a supremum over tests $\varphi : \mathcal{R} \rightarrow [0, 1]$ and the induced test $\psi(x) := \mathbb{E}[\varphi(R) \mid X = x]$. Then $|\mathbb{E}_{PK}[\varphi] - \mathbb{E}_{QK}[\varphi]| = |\mathbb{E}_P[\psi] - \mathbb{E}_Q[\psi]| \leq \text{TV}(P, Q) \|\psi\|_{\text{Lip}}$, and $\|\psi\|_{\text{Lip}} \leq \eta(K)$ by definition of $\eta(K)$. $\square$

Interpretation. If you publish a receipt field $R = g(X)$ with *no randomization*, then worst-case $\eta(K) = 1$ and the field can leak as much as the raw state. To make $\eta(K)$ small, the release must add noise or otherwise ensure that different internal states induce similar laws for the published object.

2 Local DP implies a clean TV cap

Local differential privacy is a natural way to randomize a published receipt field: it is an *interface-level* promise that does not depend on the deployment distribution.

Definition 2 (ε -local DP). *A channel K is ε -local-DP if for all x, x' and measurable $A \subseteq \mathcal{R}$,*

$$K(A \mid x) \leq e^\varepsilon K(A \mid x').$$

Proposition 1 (TV cap for ε -LDP channels). *If K is ε -local-DP, then*

$$\eta(K) \leq \frac{e^\varepsilon - 1}{e^\varepsilon + 1} = \tanh(\varepsilon/2).$$

Proof. Fix x, x' and let $P = K(\cdot \mid x)$, $Q = K(\cdot \mid x')$. By ε -LDP, the likelihood ratio satisfies $dP/dQ \in [e^{-\varepsilon}, e^\varepsilon]$ wherever densities exist (and the analogous statement holds in the purely discrete case). The worst-case TV under a bounded likelihood ratio is attained by a two-point construction and equals $(\alpha - 1)/(\alpha + 1)$ where $\alpha = \sup dP/dQ$. Setting $\alpha = e^\varepsilon$ yields the claim. $\square$

Remark 1 (Why this is useful). *Many papers in this series already bound leakage from the protocol channel. Proposition 1 bounds leakage from the published certificate/receipt channel itself. Operationally: to publish an artifact field, pick an ε for that field and record it in the receipt; then the field contributes at most $\tanh(\varepsilon/2)$ TV in the worst case.*

3 Composing multiple receipt fields

Receipts are typically tuples: $R = (R_1, \dots, R_k)$. Let K_i be the channel for field i .

Proposition 2 (Union-style bound). *If the receipt fields are generated conditionally independently given X (i.e., the channel factorizes as $K = \prod_i K_i$), then*

$$\eta(K) \leq 1 - \prod_{i=1}^k (1 - \eta(K_i)) \leq \sum_{i=1}^k \eta(K_i).$$

Proof. For fixed x, x' , couple each field channel so that $R_i = R'_i$ with probability at least $1 - \text{TV}(K_i(\cdot \mid x), K_i(\cdot \mid x'))$. Under conditional independence, the product coupling makes all fields equal with probability at least $\prod_i (1 - \text{TV}_i)$. Thus the TV of the product laws is at most the probability that at least one field differs, giving the first bound. The second is $1 - \prod_i (1 - a_i) \leq \sum_i a_i$. $\square$

Practical guideline. Keep receipt fields *few* and *separable*. If a single derived field depends on many raw internals, its effective contraction coefficient can be close to 1 unless you explicitly add noise.

4 Examples for publishable anonymity artifacts

(1) Publishing bucketed tail summaries. Suppose a system wants to publish a tail indicator (e.g., whether a latency exceeded a threshold) for monitoring. Publishing a raw Bernoulli outcome can leak heavily if it is correlated with the secret. Applying randomized response with LDP parameter ε yields an explicit $\tanh(\varepsilon/2)$ TV cap for that field.

(2) Publishing histogram sketches. If a paper wants to publish a small sketch of an empirical distribution (e.g., a congestion witness histogram or a destination-trace bucket count), then using an LDP mechanism per bin (or a private count sketch) yields an explicit per-field TV bound. Proposition 2 then gives a simple end-to-end cap.

(3) Receipt design in the series. The release-receipt paper should record *receipt-field privacy knobs* (e.g., LDP parameters or explicit redaction policies) as first-class items in the exported receipt. The closed-view auditing paper should treat published canary and monitoring summaries as receipt fields and budget them similarly.

5 What this note does *not* do

This note does not choose an optimal redaction for any specific system. It provides a small, reusable interface: if you can bound (or choose) a contraction coefficient for a published artifact field, you can budget the receipt channel alongside the protocol channel.

References

- [1] Anonymous. Artifact and Disclosure Policy for the One-True-Archive (Source-Only Public Release). Series synthesis note (Synthesis 6), 2026. (This archive.)
- [2] Anonymous. Release Property Ledger Receipts (Commitments, Redaction, and Verifiability). Release and Destination series Paper A, 2026. (This archive.)
- [3] Anonymous. ABOM and OINL for Anonymity Claims. Anonymity series Paper C, 2026. (This archive.)